

Person Detection, Tracking, and Multi-Head Action Recognition on TITAN: Initial Results

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1 Introduction

We study an end-to-end perception pipeline for urban scenes that detects and tracks people in video and then predicts fine-grained action categories for each tracked individual. The detector is a YOLO11 model configured for a single class (*person*), coupled with ByteTrack for online multi-object tracking. Cropped person tubes feed a multi-head 3D CNN (R3D-18, Kinetics-pretrained) that predicts five mutually exclusive action heads from the TITAN dataset (*atomic*, *simple-context*, *complex-context*, *communicative*, *transportive*).

For this milestone we focus on (i) quantifying how well YOLO11 can localize pedestrians on TITAN, and (ii) evaluating the action recognizer on a 20% tube subset with window length $T=8$ and stride $S=4$. We report head-wise mAP and accuracy, and give qualitative examples. The main limitation we identify is severe head-wise class imbalance, especially for complex, communicative, and transportive actions.

2 Problem and Applications

Urban-driving perception systems must reason not only about *where* people are, but also *what* they are doing over time. Anticipating near-future human motion and intent—e.g., walking vs. running vs. entering a vehicle—is crucial for safe navigation and interaction. Our goal is therefore to (1) detect and track people across frames and (2) infer their actions from short spatio-temporal tubes, producing per-person predictions for the five TITAN action heads.

3 Relationship to Course Content

This project touches several core topics in AER1515: appearance-based detection, multi-object tracking (MOT), spatio-temporal representation learning, and multi-task classification. We extend beyond static image recognition into temporal reasoning using 3D CNNs and uncertainty-weighted multi-head losses, and we evaluate using detection-style mAP and classification confusion matrices, in line with standard robotic perception practice.

4 Dataset

We use the Honda TITAN dataset [1], an urban-driving video corpus with synchronized image frames and per-object annotations sampled at 10Hz. We follow the official train/val/test

split of clips summarized in Table 1 and restrict supervision to objects labeled as *person* and to the five person-related action families used by our heads, discarding all vehicle-centric attributes.

Raw TITAN scale. Table 1 reports the number of clips, annotated frames, and person bounding boxes per split for the person subset used in all experiments.

Split	#clips	#frames	#person boxes
Train	400	40,808	221,760
Val	200	21,398	119,395
Test	100	12,371	58,155
Total	700	74,577	399,310

Table 1: Person-scale of TITAN used in our experiments.

Detection splits for YOLO11. For person detection we convert TITAN person boxes into YOLO format aligned with the same clip partition. The full (100%) detection split and a 10% subset (constructed from the first 10% of clip IDs per split) are summarized in Table 2.

Split	#images		#person boxes	
	100%	10%	100%	10%
Train	40,032	4,396	221,760	30,699
Val	21,064	3,134	119,395	21,875
Test	12,175	1,429	58,155	10,416
Total	73,271	8,959	399,310	62,990

Table 2: Scale of YOLO11 detection splits for TITAN.

Action-recognition tubes and label distribution. For action recognition we build person tubes by sliding fixed-length windows over each ground-truth track and cropping to the person box. On the 20% clip subset used for initial results, this yields 11,319 / 7,978 / 4,119 tubes for train/val/test, respectively, with $(T, S) = (8, 4)$. Each tube contributes one label per head, so each head sees 23,416 labels.

Table 3 aggregates the label distribution across train+val+test. The atomic head is almost fully annotated (very few none labels), whereas the other heads are dominated by none, indicating that many clips contain no

explicit contextual, communicative, or transportive action for the tracked person.

Head	Total	#none	#non-none	%none
Atomic	23,416	48	23,368	0.20
Simple-context	23,416	16,397	7,019	70.0
Complex-context	23,416	23,227	189	99.2
Communicative	23,416	20,896	2,520	89.2
Transportive	23,416	22,808	608	97.4

Table 3: Label distribution across heads for the 20% tube subset (train+val+test).

At a finer level, the atomic head is dominated by common motions such as *walking* and *standing*, with comparatively few instances of more dynamic or rare actions (e.g., *running*, *squatting*, *jumping*). For the simple-context head, *none* and *walking on the road* cover most labels, while door/building interactions and biking/motorcycling are sparse. Complex-context, communicative, and transportive heads contain only a few hundred positive labels each, which directly contributes to the low mAP but high accuracy observed for these heads.

5 Methodology

5.1 Detection and Tracking

Detector. We experimented with three YOLO11 configurations for single-class person detection, all trained or evaluated on a MacBook Pro with an Apple M4 Max and 48 GB of unified memory using the `mps` backend.

- **YOLO11n-full:** YOLO11n fine-tuned on 100% of the TITAN person detection split for one epoch.
- **YOLO11n-0.1:** YOLO11n fine-tuned on the 10% subset with a full training schedule, evaluated both with and without test-time augmentation (TTA).
- **YOLO11L:** YOLO11L evaluated off-the-shelf with COCO-pretrained weights (no TITAN fine-tuning).

These experiments are used to understand how well an off-the-shelf detector can localize persons on TITAN and how much benefit comes from even modest fine-tuning.

The detection results show that a properly trained YOLO11 detector on the full TITAN person split would likely deliver near-perfect pedestrian boxes. However, to cleanly evaluate the action recognizer at this milestone, we *do not* feed detector boxes downstream. All tubes for action recognition are built from ground-truth TITAN person boxes.

Tracker. Our target online pipeline uses ByteTrack to associate per-frame person detections into tracklets. For each clip, the tracker would output a MOT-style text file of the form

frame,id,x,y,w,h,1,-1,-1,-1,

and an MP4 visualization with overlaid track IDs. In the current milestone, we bypass tracking and construct “oracle”

tubes directly from ground-truth person tracks, keeping the ByteTrack setup in reserve for the next stage when we switch to predicted boxes.

5.2 Action Recognition from Person Tubes

Given a ground-truth person track and its per-frame bounding boxes, we extract fixed-length tubes by cropping a window $\{I_t\}_{t=1}^T$ around the box (top-left $xywh$) and resizing each crop to 224×224 . We slide this window along the track with stride S to generate tube samples; in this milestone we train on a 20% clip subset with $(T, S) = (8, 4)$.

Backbone. Our action recognizer is an R3D-18 network from `torchvision`, pretrained on Kinetics-400. We replace the final fully connected layer by **five** linear classification heads, one per TITAN action family: *atomic*, *simple-context*, *complex-context*, *communicative*, and *transportive*. Each head predicts a single mutually exclusive label, including a *none* class.

Loss. We use homoscedastic uncertainty weighting across heads combined with label smoothing. For head h we maintain a learnable log-variance s_h and optimize

$$\mathcal{L} = \sum_h \exp(-s_h) \text{CE}_\epsilon(\hat{\mathbf{y}}_h, \mathbf{y}_h) + s_h,$$

with label-smoothing parameter $\epsilon = 0.05$. This lets the model down-weight easier heads and prevents over-confident predictions.

Optimization. Training is performed on the M4 Max using the `mps` backend. We use SGD with Nesterov momentum (0.9), initial learning rate 7×10^{-4} , weight decay 10^{-4} , batch size 24, and 4 dataloader workers. The learning rate follows a cosine schedule over a 40-epoch budget with a 3-epoch warmup. Gradients are clipped to a norm of 1.0. To partially mitigate the long-tail label distribution, we enable a rare-class sampler with repeat factor 1.5, which oversamples tubes containing rare labels. We apply early stopping on validation macro-mAP with patience 8; in this run training stopped at epoch 32, and we select the checkpoint with the best validation macro-mAP for test-time evaluation.

Evaluation. For each head we compute per-class average precision (AP) and head mAP (excluding *none*), as well as head accuracy and macro-averaged mAP/accuracy across heads. We also export precision-recall curves and confusion matrices (CSV and PNG) for analysis, and generate qualitative visualizations. These include single-tube examples with labels near the box, and a multi-person frame where all persons in a scene are shown with colour-coded boxes and a legend listing their ground-truth and predicted actions.

6 Results

6.1 Detection and Tracking

Quantitative detection results. We benchmark the three YOLO11 configurations on the TITAN person test set (1,429 images, 10,416 person instances). Table 4 reports box mAP metrics; Box-F1 scores are 0.987, 0.950, 0.969, and 0.969 for YOLO11n-full, YOLO11n-0.1 (w/o TTA), YOLO11n-0.1 (w/ TTA), and YOLO11L, respectively.

Model	Data	mAP ₅₀	mAP _{50:95}
YOLO11n-full	100% (1 epoch)	0.987	0.840
YOLO11n-0.1 (w/o TTA)	10% (merged)	0.952	0.793
YOLO11n-0.1 (w/ TTA)	10% (merged)	0.970	0.810
YOLO11L	100% (0 epoch)	0.970	0.838

Table 4: Person detection performance on the TITAN test set. All three setups achieve strong person localization; YOLO11n-full (100%) obtains the best overall mAP.

Even a single epoch of fine-tuning YOLO11n on the full TITAN detection split yields the highest mAP, suggesting that dataset coverage is more important than longer training on a small subset. The larger YOLO11L backbone, even without TITAN-specific fine-tuning, nearly matches YOLO11n-full. YOLO11n trained on only 10% of the data lags behind but recovers a large fraction of the gap when evaluated with TTA.

Qualitative comparison. Figure 1 shows a representative frame overlaid with ground-truth boxes and detections from YOLO11n-full, YOLO11n-0.1, and YOLO11L. YOLO11n-full and YOLO11L both localize small or partially occluded pedestrians robustly, whereas the 10% YOLO11n model occasionally misses distant persons or produces slightly misaligned boxes, consistent with its lower recall.

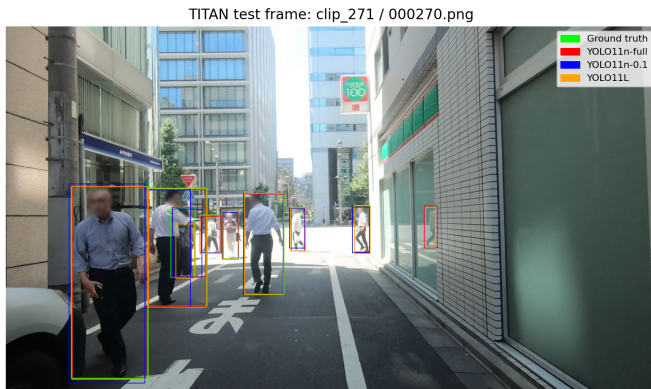


Figure 1: Qualitative comparison of ground-truth person boxes and detections from YOLO11n-full (100%), YOLO11n-0.1 (10%), and YOLO11L on the same TITAN frame.

Downstream usage. Given these results, we expect that a fully trained YOLO11n or YOLO11L on 100% TITAN could

provide high-quality person boxes for action recognition. In this milestone we instead use ground-truth boxes to isolate the behaviour of the multi-head action recognizer; integrating detector outputs into the tube extraction pipeline is left to the final project stage.

6.2 Action Recognition

We train the R3D-18 backbone with five action heads on the 20% TITAN tube subset ($T=8$, $S=4$, image size 224^2) and evaluate on the held-out test tubes.

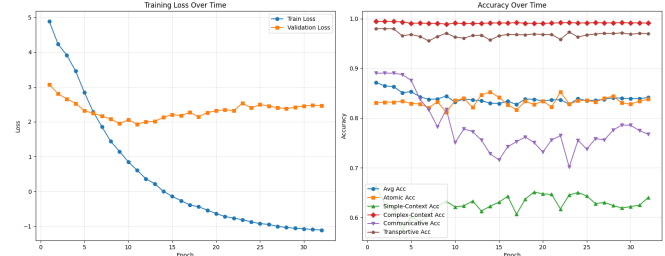


Figure 2: Training and validation loss (left) and head-wise accuracy (right) over epochs for the R3D-18 action recognizer.

Figure 2 shows that training loss decreases steadily, while validation loss plateaus around epoch 10 and then slowly rises. This indicates that the model fits the training data well but begins to overfit the dominant classes once validation performance saturates. The accuracy curves show that atomic, complex-context, and transportive heads remain consistently high, while simple-context and communicative heads are both harder and more unstable, reflecting their more diverse and imbalanced label distributions.

Head-wise test performance. Table 5 reports test accuracy and mAP for each head. We exclude the `none` class from mAP and compute macro averages across heads.

Head	Accuracy	mAP (excl. none)
Atomic	0.774	0.360
Simple-context	0.566	0.158
Complex-context	0.986	0.005
Communicative	0.855	0.079
Transportive	0.921	0.015
Macro average	0.821	0.123

Table 5: Per-head test accuracy and mAP. mAP is computed over non-`none` classes within each head and then averaged across heads.

Overall, the macro-averaged test accuracy is 0.82 with a macro mAP of 0.12. Atomic actions achieve the strongest balance of accuracy and mAP; the model reliably recognizes common motions but struggles on rare ones. Simple-context and communicative heads are clearly impacted by their long-tail distributions, while complex-context and transportive

heads achieve very high accuracy but almost zero mAP, meaning the model mostly predicts `none` correctly and seldom recovers the rare positive actions.

Per-class AP highlights. Per-class AP values illustrate this pattern more explicitly. For the atomic head, frequent actions such as *walking* and *standing* reach high AP (around 0.9 and 0.55, respectively), and *sitting* is also well recognized. In contrast, rarer actions such as *bending*, *running*, and *squatting* have near-zero AP.

For simple-context, the model performs reasonably on *biking* and *walking on the road*, but AP for door- and building-related interactions (*entering/exiting*, *opening/closing*), as well as *motorcycling*, is essentially zero. Complex-context AP is extremely low for both *loading* and *unloading*. Communicative actions achieve modest AP, with *talking in group* better captured than *talking on phone*. Transportive actions (*pulling*, *pushing*, *carrying*) remain very challenging.

Qualitative tube-level predictions. Figure 3 shows a representative test frame where all annotated persons are overlaid with colour-coded bounding boxes. On the right, a legend lists each person together with their ground-truth and predicted actions (for heads with non-`none` labels), using the same colour as the box. The model generally predicts atomic *walking/standing* correctly, and is often correct for communicative actions like *talking on phone* or *talking in group*. However, some pedestrians engaged in phone use are still mapped to `none`, and no complex loading/unloading behaviour appears in this frame, highlighting the scarcity of those classes in typical scenes.



Figure 3: Qualitative example from the test set. Each annotated person has a colour-coded bounding box; the legend lists ground-truth and predicted actions (non-`none` heads only) for each person using the same colour.

7 Future Work and Timeline

Given the remaining four weeks (two for the video demo, two for the final report), we focus on small but targeted improvements and additional analysis rather than major architectural changes.

Short-term improvements.

- **Temporal extent ablation:** Train a second R3D-18 model with longer tubes ($T=16$) on the same 20% subset to quantify the benefit of longer temporal context on atomic and simple-context mAP.
- **Imbalance correction:** Rerun training with the a better tuned rare-class sampler and compare per-head mAP/accuracy to isolate the effect of class imbalance correction on the tails.
- **Qualitative coverage:** Collect a small set (5–10) of additional multi-person frames emphasizing rare communicative and transportive actions for use in the video and final report.

Milestones.

- **By video deadline:** Finalize detector and action-recognizer checkpoints; generate qualitative detection and action clips; and edit a 3–5 minute demo video combining detection/tracking and tube-level predictions.
- **By final report):** Integrate imbalance correction results, update figures/tables (including PR curves and key confusion matrices), and expand the error analysis section on rare actions before submitting the final report.

References

- [1] S. Malla, B. Dariush, and C. Choi, “TITAN: Future Forecast using Action Priors,” in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 11186–11196.